

Supplementary Material: Molecular Docking: A Machine-Checked Theory of Exact Resolution, Complexity, and Thermodynamic Cost

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April 20, 2026

1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AC1	ClaimClosure.AtomicCircuitExports.AC1 paper4/DecisionQuotient/ClaimClosure.lean	AC3	ClaimClosure.AtomicCircuitExports.AC3 paper4/DecisionQuotient/ClaimClosure.lean
AC4	ClaimClosure.AtomicCircuitExports.AC4 paper4/DecisionQuotient/ClaimClosure.lean	BA1	Physics.BoundedAcquisition.BoundedRegion paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA2	Physics.BoundedAcquisition.acquisition_rate_ bound paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA3	Physics.BoundedAcquisition.acquisitions_are_ transitions paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA4	Physics.BoundedAcquisition.one_bit_per_ transition paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA5	Physics.BoundedAcquisition.resolution_reads_ sufficient paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA6	Physics.BoundedAcquisition.srank_le_ resolution_bits paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA7	Physics.BoundedAcquisition.energy_ge_srank_ cost paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA8	Physics.BoundedAcquisition.srank_one_energy_ minimum paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA10	Physics.BoundedAcquisition.counting_gap_ theorem paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
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CV9	Physics.Conversation.clamp_event_implies_ positive_energy paper4/DecisionQuotient/Physics/ Conversation.lean	DT22	Physics.DecisionTime.substrate_step_realizes_ decision_event paper4/DecisionQuotient/Physics/ DecisionTime.lean
DT23	Physics.DecisionTime.substrate_step_is_time_ unit paper4/DecisionQuotient/Physics/ DecisionTime.lean	DT24	Physics.DecisionTime.time_unit_law_substrate_ invariant paper4/DecisionQuotient/Physics/ DecisionTime.lean
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L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean	L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean
L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean	L51	ssot_srank_one Leverage/BridgeToDQ.lean
L52	succ_le_two_pow Leverage/BridgeToDQ.lean	L53	sufficiency_conp_hard
L54	thermodynamic_selection Leverage/BridgeToDQ.lean	L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean
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L58	Leverage.ColumnComplexityBridge. zeroIdentityDebt_tieBrokenArgmin_of_uniform_ argmin_relation_bound Leverage/ColumnComplexityBridge.lean	L60	Leverage.rattle_constraintObservations_card Leverage/BridgeToDQ.lean
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L87	Leverage.boundedActions_polynomialTime Leverage/DockingTheoryBridge.lean	L88	Leverage.gridMDState_sufficient_erase_ irrelevant Leverage/DockingTheoryBridge.lean
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L133	Leverage.mechanochemical_coupling_gap Leverage/DockingTheoryBridge.lean	L134	Leverage.hierarchical_srank_bound Leverage/DockingTheoryBridge.lean

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L422	Leverage.binding_free_energy_floor_tight_iff_ residualEntropy_zero Leverage/DockingTheoryBridge.lean	L423	Leverage.quotientTrajectory_crooks_standard_ form_of_work_calibration Leverage/DockingTheoryBridge.lean
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L426	Leverage.hopfield_ninio_dissipation_overhead_ lower_bound Leverage/DockingTheoryBridge.lean	L427	Leverage.hopfield_ninio_fault_tolerant_rank_ overhead_lower_bound Leverage/DockingTheoryBridge.lean
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L453	Leverage.collapsedRankProduct_universal Leverage/DockingTheoryBridge.lean	L454	Leverage.collapsedRankCoproduct_universal Leverage/DockingTheoryBridge.lean
L455	Leverage.collapsedRankTensor_map Leverage/DockingTheoryBridge.lean	L456	Leverage.collapsedRankStructuralFunctor_mono Leverage/DockingTheoryBridge.lean
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L489	Leverage.decisionEncodingTransport_energyLowerBound_eq Leverage/BridgeToDQ.lean	L490	Leverage.OperationKernelSchema.abstraction_has_unique_factorization_to_kernel_quotient Leverage/BridgeToDQ.lean
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L513	Leverage.measureKernel_transport_endpoint Leverage/BridgeToDQ.lean	L514	Leverage.measurableDeterministicKernelSemigroup Leverage/BridgeToDQ.lean
L515	Leverage.measurableDeterministicInvariantLayer Leverage/BridgeToDQ.lean	L516	Leverage.measurableDeterministic_scale_stationary_of_invariant Leverage/BridgeToDQ.lean
L517	Leverage.MeasureKernelSemigroup.stationary_kernelPow Leverage/BridgeToDQ.lean	L518	Leverage.MeasureKernelSemigroup.detailedBalance_stationary_kernelPow Leverage/BridgeToDQ.lean

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L519	Leverage.MeasureKernelSemigroupHom.map_ kernelPow Leverage/BridgeToDQ.lean	L520	Leverage.MeasureKernelSemigroupHom.stationary_ transport_kernelPow Leverage/BridgeToDQ.lean
L521	Leverage.DetailedBalanceSemigroupHom. stationary_transport_kernelPow_of_ detailedBalance Leverage/BridgeToDQ.lean	L522	Leverage.MeasureKernelSemigroup.ScaleFlow. kernelAt_eq_kernelPow_of_one Leverage/BridgeToDQ.lean
L523	Leverage.MeasureKernelSemigroup.ScaleFlow. stationary_of_detailedBalance_at_one Leverage/BridgeToDQ.lean	L524	Leverage.DetailedBalanceSemigroupHom. stationary_transport_scaleFlow_of_ detailedBalance_at_one Leverage/BridgeToDQ.lean
L525	Leverage.PathSpaceCrooksModel.logRatio_ kernelPow_eq_zero_of_stationary Leverage/BridgeToDQ.lean	L526	Leverage.PathSpaceCrooksModel.logRatio_ kernelPow_eq_zero_of_detailedBalance Leverage/BridgeToDQ.lean
L527	Leverage.PathSpaceCrooksModel.logRatio_scale_ eq_zero_of_detailedBalance_at_one Leverage/BridgeToDQ.lean	L528	Leverage.PathSpaceCrooksTransport.logRatio_ transport_kernelPow Leverage/BridgeToDQ.lean
L529	Leverage.PathSpaceCrooksTransport.logRatio_ kernelPow_eq_zero_of_source_detailedBalance Leverage/BridgeToDQ.lean	L530	Leverage.measurableTransitionKernelSemigroup Leverage/BridgeToDQ.lean
L531	Leverage.measurableTransition_scale_ stationary_of_detailedBalance_at_one Leverage/BridgeToDQ.lean	L532	Leverage.measurableTransition_pathSpaceCrooks_ zero_of_detailedBalance_at_one Leverage/BridgeToDQ.lean
L533	Leverage.PathSpaceCrooksModel.logRatio_ scaleFlow_eq_zero_of_transported_ detailedBalance_at_one Leverage/BridgeToDQ.lean	L534	Leverage.PathSpaceCrooksTransport.logRatio_ scaleFlow_eq_zero_of_source_detailedBalance_ at_one Leverage/BridgeToDQ.lean
L535	Leverage.deterministicToTransitionSemigroupHom Leverage/BridgeToDQ.lean	L536	Leverage.deterministicToTransition_stationary_ kernelPow Leverage/BridgeToDQ.lean
L537	Leverage.PathSpaceJarzynskiModel. expNegLogRatioExpectation_kernelPow_eq_one_ of_stationary Leverage/BridgeToDQ.lean	L538	Leverage.PathSpaceJarzynskiModel. expNegLogRatioExpectation_kernelPow_eq_one_ of_detailedBalance Leverage/BridgeToDQ.lean
L539	Leverage.PathSpaceJarzynskiModel. expNegLogRatioExpectation_scale_eq_one_of_ detailedBalance_at_one Leverage/BridgeToDQ.lean	L540	Leverage.PathSpaceJarzynskiTransport. expNegLogRatioExpectation_transport_kernelPow Leverage/BridgeToDQ.lean
L541	Leverage.PathSpaceJarzynskiTransport. expNegLogRatioExpectation_kernelPow_eq_one_ of_source_detailedBalance Leverage/BridgeToDQ.lean	L542	Leverage.measurableTransition_ pathSpaceJarzynski_one_of_detailedBalance_ at_one Leverage/BridgeToDQ.lean
L543	Leverage.PathSpaceExpectationModel. pathIntegral_expNegLogRatio_eq_one_of_ logRatio_eq_zero Leverage/BridgeToDQ.lean	L544	Leverage.PathSpaceExpectationModel. pathIntegral_expNegLogRatio_kernelPow_eq_ one_of_stationary Leverage/BridgeToDQ.lean
L545	Leverage.PathSpaceExpectationModel. pathIntegral_expNegLogRatio_kernelPow_eq_ one_of_detailedBalance Leverage/BridgeToDQ.lean	L546	Leverage.PathSpaceExpectationModel. pathIntegral_expNegLogRatio_scale_eq_one_ of_detailedBalance_at_one Leverage/BridgeToDQ.lean
L547	Leverage.PathSpaceJarzynskiTransport. pathIntegral_expNegLogRatio_transport_ kernelPow Leverage/BridgeToDQ.lean	L548	Leverage.PathSpaceJarzynskiTransport. pathIntegral_expNegLogRatio_kernelPow_eq_ one_of_source_detailedBalance Leverage/BridgeToDQ.lean
L549	Leverage.measurableTransition_ pathSpaceJarzynski_pathIntegral_one_of_ detailedBalance_at_one Leverage/BridgeToDQ.lean	L550	Leverage.MeasureKernelQuotientCalculus. stationary_descends Leverage/BridgeToDQ.lean
L551	Leverage.MeasureKernelSemigroupQuotientCalculus. stationary_kernelPow_descends Leverage/BridgeToDQ.lean	L552	Leverage.DetailedBalanceSemigroupQuotientCalculus. stationary_kernelPow_descends_of_ detailedBalance Leverage/BridgeToDQ.lean

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L553	Leverage.DetailedBalanceSemigroupQuotientCalculus. stationary_scaleFlow_descends_of_ detailedBalance_at_one Leverage/BridgeToDQ.lean	L554	Leverage.KernelSemigroupEndomorphism.map_ kernelPow Leverage/BridgeToDQ.lean
L555	Leverage.QuotientKernelRGCompatibility. commutes_kernelPow Leverage/BridgeToDQ.lean	L556	Leverage.DetailedBalanceQuotientKernelRGCompatibility. stationary_targetRG_kernelPow_of_source_ detailedBalance Leverage/BridgeToDQ.lean
L557	Leverage.DetailedBalanceQuotientKernelRGCompatibility. endpoint Leverage/BridgeToDQ.lean	L558	Leverage.MeasureKernelSemigroupHom.stationary_ transport_kernelPow_as_quotient_instance Leverage/BridgeToDQ.lean
L559	Leverage.DetailedBalanceSemigroupHom. stationary_transport_kernelPow_as_quotient_ instance Leverage/BridgeToDQ.lean	L560	Leverage.PathSpaceExpectationModel. pathIntegral_scaleFlow_eq_one_of_transported_ detailedBalance_at_one Leverage/BridgeToDQ.lean
L562	Leverage.PathSpaceProcessTransport. expNegLogRatioExpectation_transport_kernelPow Leverage/BridgeToDQ.lean	L563	Leverage.PathSpaceProcessTransport. pathIntegral_transport_kernelPow Leverage/BridgeToDQ.lean
L564	Leverage.PathSpaceProcessTransport. pathIntegral_kernelPow_eq_one_of_source_ detailedBalance Leverage/BridgeToDQ.lean	L565	Leverage.measurableTransitionFinitePathMeasure_ zero Leverage/BridgeToDQ.lean
L566	Leverage.measurableTransitionFinitePathMeasure_ succ Leverage/BridgeToDQ.lean	L567	Leverage.measurableTransition_finiteHorizon_ pathIntegral_scale_eq_one_of_detailedBalance_ at_one Leverage/BridgeToDQ.lean
L568	Leverage.measurableTransition_finiteHorizon_ processTransport_kernelPow_pathIntegral_eq_ one_of_source_detailedBalance Leverage/BridgeToDQ.lean	L569	Leverage.measurableTransition_finiteHorizon_ pathIntegral_scale_eq_one_of_transported_ detailedBalance_at_one Leverage/BridgeToDQ.lean
L570	Leverage.MeasurableTransitionFiniteHorizonProjectiveConsistency. marginal_eq Leverage/BridgeToDQ.lean	L571	Leverage.MeasurableTransitionFiniteHorizonProjectiveConsistency. marginal_kernelPow_eq Leverage/BridgeToDQ.lean
L572	Leverage.MeasurableTransitionFiniteHorizonProjectiveConsistency. marginal_scaleFlow_eq Leverage/BridgeToDQ.lean	L573	Leverage.MeasurableTransitionKolmogorovExtension. marginal_recovery Leverage/BridgeToDQ.lean
L574	Leverage.MeasurableTransitionKolmogorovExtension. marginal_pathIntegral_scale_eq_one_of_ detailedBalance_at_one Leverage/BridgeToDQ.lean	L575	Leverage.MeasurableTransitionKolmogorovExtension. marginal_pathIntegral_scale_eq_one_of_ transported_detailedBalance_at_one Leverage/BridgeToDQ.lean
L576	Leverage.measurableTransitionFiniteHorizon_ projectiveConsistency_instance Leverage/BridgeToDQ.lean	L577	Leverage.ToleranceCollapseProfile. collapseCount_eq_formula Leverage/DockingTheoryBridge.lean
L578	Leverage.LJPocketToleranceCollapseProfile. collapseCount_eq_formula Leverage/DockingTheoryBridge.lean	L579	Leverage.CompositeConstraintDecisionProcedure. srnk_eq_effectiveDOF Leverage/DockingTheoryBridge.lean
L580	Leverage.MolecularDynamicsNonequilibriumCalibration. crooks_standard_form Leverage/DockingTheoryBridge.lean	L581	Leverage.ContinuousTimeKernelSemigroup. discreteScaleFlow Leverage/DockingTheoryBridge.lean
L582	Leverage.GeneralObservationFisherInterface. totalFisher_eq_srnk Leverage/DockingTheoryBridge.lean	L583	Leverage.GeneralObservationFisherInterface. relevant_iff_fisher_eq_one Leverage/DockingTheoryBridge.lean
L584	Leverage.AlphabetRichnessInterface.srnk_ge_ of_numOptClasses_ge_pow Leverage/DockingTheoryBridge.lean	L585	Leverage.VCStyleRichnessWitness.vcDim_le_srnk Leverage/DockingTheoryBridge.lean
L586	Leverage.inverseDesign_atomistic_realization Leverage/DockingTheoryBridge.lean	L587	Leverage.ContinuousStatePathExtensionInterface. measurable_evalAt_pair Leverage/DockingTheoryBridge.lean
L588	Leverage.MeasurableTransitionFinitePath. measurable_truncate Leverage/BridgeToDQ.lean	L589	Leverage.ComposedHamiltonian_Architecture_ instance Leverage/DockingTheoryBridge.lean
L590	Leverage.ComposedHamiltonian_Lipschitz_bound Leverage/DockingTheoryBridge.lean	L591	Leverage.Langevin_satisfies_DetailedBalance Leverage/DockingTheoryBridge.lean

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L592	Leverage.Langevin_Discretization_to_MCMC Leverage/DockingTheoryBridge.lean	L593	Leverage.SpinHalf_two_states Leverage/DockingTheoryBridge.lean
L594	Leverage.SpinHalf_CanonicalDP_instance Leverage/DockingTheoryBridge.lean	L595	Leverage.SpinHalf_Decoherence_Cost Leverage/DockingTheoryBridge.lean
L596	Leverage.TopLevel_Computable Leverage/DockingTheoryBridge.lean	L597	Leverage.NoisyPartialObservationChannel. totalFisher_eq_srank Leverage/DockingTheoryBridge.lean
L598	Leverage.NoisyPartialObservationChannel. relevant_iff_debiasedFisher_eq_one Leverage/DockingTheoryBridge.lean	L599	Leverage.langevin_solution_exists_unique Leverage/DockingTheoryBridge.lean
L600	Leverage.langevin_boltzmann_invariant Leverage/DockingTheoryBridge.lean	L601	Leverage.langevin_ergodic Leverage/DockingTheoryBridge.lean
L602	Leverage.eulerMaruyama_strong_error_bound Leverage/DockingTheoryBridge.lean	L603	Leverage.eulerMaruyama_weak_error_bound Leverage/DockingTheoryBridge.lean
L604	Leverage.concreteLangevin_to_interface Leverage/DockingTheoryBridge.lean	L605	Leverage.concreteComposedHamiltonian_ Architecture_instance Leverage/DockingTheoryBridge.lean
L606	Leverage.concreteComposedHamiltonian_ lipschitz_bound Leverage/DockingTheoryBridge.lean	L607	Leverage.concreteComposedHamiltonian_ halfGapTransport Leverage/DockingTheoryBridge.lean
L608	Leverage.chemical_augmented_opt_eq_projection Leverage/DockingTheoryBridge.lean	L609	Leverage.chemical_augmented_srank_eq_of_ witness Leverage/DockingTheoryBridge.lean
L610	Leverage.conformationalEnsemble_population_ normalized Leverage/DockingTheoryBridge.lean	L611	Leverage.ensemble_to_docking_transport Leverage/DockingTheoryBridge.lean
L612	Leverage.inducedFit_rank_transport Leverage/DockingTheoryBridge.lean	L613	Leverage.kinetic_onRate_bound Leverage/DockingTheoryBridge.lean
L614	Leverage.kinetic_residence_eq_inverse_koff Leverage/DockingTheoryBridge.lean	L615	Leverage.kinetic_pathway_population_normalized Leverage/DockingTheoryBridge.lean
L616	Leverage.kinetic_observable_bundle Leverage/DockingTheoryBridge.lean	L617	Leverage.legacy_constructive_equiv Leverage/DockingTheoryBridge.lean
L618	Leverage.constructive_deprecation_ready Leverage/DockingTheoryBridge.lean	L619	Leverage.langevin_endpoints_of_analysis_ assumptions Leverage/DockingTheoryBridge.lean
L620	Leverage.concreteComposedHamiltonian_zero_ shell_constants Leverage/DockingTheoryBridge.lean	L621	Leverage.biomolecularReference_zero_shell_ constants Leverage/DockingTheoryBridge.lean
L622	Leverage.concreteComposedHamiltonian_global_ zero_lipschitz Leverage/DockingTheoryBridge.lean	L623	Leverage.chemical_augmented_opt_eq_projection_ of_binding_problem Leverage/DockingTheoryBridge.lean
L624	Leverage.chemical_augmented_srank_eq_of_ binding_problem Leverage/DockingTheoryBridge.lean	L625	Leverage.ensemble_to_binding_problem_transport Leverage/DockingTheoryBridge.lean
L626	Leverage.inducedFit_rank_transport_to_binding_ problem Leverage/DockingTheoryBridge.lean	L627	Leverage.docking_kinetic_observable_bundle Leverage/DockingTheoryBridge.lean
L628	Leverage.constructive_replaces_legacy_output_ fields Leverage/DockingTheoryBridge.lean	L629	Leverage.downstream_consumer_transport_of_ constructive_equiv Leverage/DockingTheoryBridge.lean
L630	Leverage.langevin_endpoints_of_first_ principles Leverage/DockingTheoryBridge.lean	L631	Leverage.concreteComposedHamiltonian_ nontrivial_lipschitz_bound Leverage/DockingTheoryBridge.lean
L632	Leverage.concreteComposedHamiltonian_ nontrivial_halfGapTransport Leverage/DockingTheoryBridge.lean	L633	Leverage.chemical_component_variation_ preserves_opt Leverage/DockingTheoryBridge.lean
L634	Leverage.oneStepConformerPopulation_sum_one Leverage/DockingTheoryBridge.lean	L635	Leverage.kinetic_observable_bundle_of_ protocol_measurements Leverage/DockingTheoryBridge.lean
L636	Leverage.fullyConstructivePipeline_ deprecation_ready Leverage/DockingTheoryBridge.lean	L637	Leverage.LangevinTransitionKernel.boltzmann_ stationary_of_detailedBalance Leverage/DockingTheoryBridge.lean

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L638	Leverage.LangevinFirstPrinciplesAssumptions. explicit_sde_conditions Leverage/DockingTheoryBridge.lean	L639	Leverage.LangevinFirstPrinciplesAssumptions. ergodic_of_dissipativity Leverage/DockingTheoryBridge.lean
L640	Leverage.concreteComposedHamiltonian_ fullState_lipschitz_bound Leverage/DockingTheoryBridge.lean	L641	Leverage.concreteComposedHamiltonian_ fullState_halfGapTransport Leverage/DockingTheoryBridge.lean
L642	Leverage.biomolecularCalibrated_fullState_ shell_constants_pos Leverage/DockingTheoryBridge.lean	L643	Leverage.chemical_component_variation_changes_ utility Leverage/DockingTheoryBridge.lean
L644	Leverage.chemical_component_variation_can_ change_opt Leverage/DockingTheoryBridge.lean	L645	Leverage.nStepConformerPopulation_transport Leverage/DockingTheoryBridge.lean
L646	Leverage.kinetic_observable_bundle_with_ confidence Leverage/DockingTheoryBridge.lean	L647	Leverage.kinetic_protocol_confidence_transport Leverage/DockingTheoryBridge.lean
L648	Leverage.constructive_only_spec_iff_legacy_ spec Leverage/DockingTheoryBridge.lean	L649	Leverage.paper4_witness_chain_to_paper3_ endpoints Leverage/DockingTheoryBridge.lean
L650	Leverage.continuous_langevin_closure_of_ measure_theoretic_assumptions Leverage/DockingTheoryBridge.lean	L651	Leverage.ContinuousLangevinMeasureClosure. stationary_all_scales Leverage/DockingTheoryBridge.lean
L652	Leverage.concreteComposedHamiltonian_pairwise_ lipschitz_bound Leverage/DockingTheoryBridge.lean	L653	Leverage.concreteComposedHamiltonian_pairwise_ halfGapTransport Leverage/DockingTheoryBridge.lean
L654	Leverage.calibratedBiophysical_changes_ utility_at_fixed_core Leverage/DockingTheoryBridge.lean	L655	Leverage.calibratedBiophysical_can_change_opt Leverage/DockingTheoryBridge.lean
L656	Leverage.EnsemblePopulationNoiseModel. observable_expectation_error_bound Leverage/DockingTheoryBridge.lean	L657	Leverage.kinetic_protocol_inference_guarantee Leverage/DockingTheoryBridge.lean
L658	Leverage.constructive_only_core_field_ consumers Leverage/DockingTheoryBridge.lean	L659	Leverage.continuous_langevin_closure_of_ito_ fokkerplanck_harris Leverage/DockingTheoryBridge.lean
L660	Leverage.concreteComposedHamiltonian_pairwise_ lipschitz_bound_of_geometric_assumptions Leverage/DockingTheoryBridge.lean	L661	Leverage.concreteComposedHamiltonian_pairwise_ halfGapTransport_of_geometric_assumptions Leverage/DockingTheoryBridge.lean
L662	Leverage.ContinuousLangevinMeasureClosure. explicit_sde_constants_of_pairwise_geometry Leverage/DockingTheoryBridge.lean	L663	Leverage.ChemicalPosteriorCalibration.robust_ protonation_utility_separation Leverage/DockingTheoryBridge.lean
L664	Leverage.ChemicalCalibrationDataset.posterior_ protonation_margin Leverage/DockingTheoryBridge.lean	L665	Leverage.kinetic_protocol_inference_guarantee_ of_concentration_identifiability Leverage/DockingTheoryBridge.lean
L666	Leverage.constructive_only_extended_field_ consumers Leverage/DockingTheoryBridge.lean	L667	Leverage.ContinuousLangevinMeasureClosure. forcefield_lipschitz_from_pairwise_geometry Leverage/DockingTheoryBridge.lean
L668	Leverage.paper4_tur_bound_of_certificate Leverage/DockingTheoryBridge.lean	L669	Leverage.paper4_velocity_verlet_shadow_ hamiltonian_bound_of_certificate Leverage/DockingTheoryBridge.lean
L670	Leverage.paper4_extended_witness_chain_to_ paper3_endpoints Leverage/DockingTheoryBridge.lean	L671	Leverage.paper4_stochastic_relevance_ conjecture_of_full_support Leverage/DockingTheoryBridge.lean
L672	Leverage.paper4_stochastic_relevance_ containment_necessary_of_nonneg Leverage/DockingTheoryBridge.lean	L673	Leverage.paper4_stochastic_relevance_ equivalence_of_nonneg_support_bridge Leverage/DockingTheoryBridge.lean
L674	Leverage.paper4_stochastic_relevance_ conjecture_of_nonneg_support_transport Leverage/DockingTheoryBridge.lean	L675	Leverage.concreteComposedHamiltonian_with_ realism_lipschitz_bound Leverage/DockingTheoryBridge.lean
L676	Leverage.concreteComposedHamiltonian_with_ realism_halfGapTransport Leverage/DockingTheoryBridge.lean	L677	Leverage.paper4_ewald_long_range_certificates Leverage/DockingTheoryBridge.lean

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L678	Leverage.FlexibleContinuumDockingLayer. fullstate_energy_lipschitz_transport Leverage/DockingTheoryBridge.lean	L679	Leverage.continuous_langevin_closure_of_ infinite_dimensional_path_measure Leverage/DockingTheoryBridge.lean
L680	Leverage.ChemicalRealDataCalibrationLayer. robust_protonation_utility_separation_with_ bias Leverage/DockingTheoryBridge.lean	L681	Leverage.kinetic_replicate_inference_bundle Leverage/DockingTheoryBridge.lean
L682	Leverage.paper4_stochastic_relevance_ conjecture_of_nonneg_primitive_dynamics Leverage/DockingTheoryBridge.lean	L683	Leverage.ConstructiveEmpiricalRealismInstantiation. finite_sample_shell_envelope Leverage/DockingTheoryBridge.lean
L684	Leverage.constructive_empirical_realism_ lipschitz_bound Leverage/DockingTheoryBridge.lean	L685	Leverage.constructive_empirical_realism_ halfGapTransport Leverage/DockingTheoryBridge.lean
L686	Leverage.continuous_langevin_closure_of_ constructive_infinite_dimensional_path_ derivation Leverage/DockingTheoryBridge.lean	L687	Leverage.continuous_langevin_closure_of_ constructive_infinite_dimensional_path_measure Leverage/DockingTheoryBridge.lean
L688	Leverage.hierarchical_chemical_realdata_ separation_of_rate Leverage/DockingTheoryBridge.lean	L689	Leverage.hierarchical_kinetic_replicate_ inference_bundle Leverage/DockingTheoryBridge.lean
L690	Leverage.paper4_stochastic_relevance_ conjecture_of_nonneg_explicit_step_dynamics Leverage/DockingTheoryBridge.lean	L691	Leverage.biomolecularRealismReferenceCalibration_ shell_values Leverage/DockingTheoryBridge.lean
L692	Leverage.biomolecularRealismReference_ constructive_lipschitz_bound Leverage/DockingTheoryBridge.lean	L693	Leverage.biomolecularRealismReference_ constructive_halfGapTransport Leverage/DockingTheoryBridge.lean
L694	Leverage.continuous_langevin_closure_of_ finite_horizon_law_derivation Leverage/DockingTheoryBridge.lean	L695	Leverage.continuous_langevin_closure_of_ finite_horizon_law_bridge Leverage/DockingTheoryBridge.lean
L696	Leverage.hierarchical_chemical_realdata_ separation_of_rate_margin Leverage/DockingTheoryBridge.lean	L697	Leverage.hierarchical_kinetic_replicate_ inference_bundle_of_rate_margins Leverage/DockingTheoryBridge.lean
L698	Leverage.paper4_support_transport_of_explicit_ step_dynamics Leverage/DockingTheoryBridge.lean	L699	Leverage.paper4_stochastic_relevance_ conjecture_of_nonneg_support_transport_of_ explicit_step_dynamics Leverage/DockingTheoryBridge.lean
L700	Leverage.biomolecularRealismReference_ correction_shell_constant_value Leverage/DockingTheoryBridge.lean	L701	Leverage.biomolecularRealismReference_ constructive_lipschitz_bound_explicit_shell Leverage/DockingTheoryBridge.lean
L702	Leverage.finite_horizon_law_bridge_projective_ and_extension_marginals Leverage/DockingTheoryBridge.lean	L703	Leverage.hierarchical_chemical_realdata_ separation_of_rate_margin_all_datasets Leverage/DockingTheoryBridge.lean
L704	Leverage.hierarchical_kinetic_replicate_ inference_bundle_of_rate_margins_all_datasets Leverage/DockingTheoryBridge.lean	L705	Leverage.HierarchicalChemicalRealDataLayer. dataset_rate_margin_of_rate_constant_bound Leverage/DockingTheoryBridge.lean
L706	Leverage.hierarchical_chemical_realdata_ separation_of_rate_constant_margin Leverage/DockingTheoryBridge.lean	L707	Leverage.hierarchical_chemical_realdata_ separation_of_rate_constant_margin_all_ datasets Leverage/DockingTheoryBridge.lean
L708	Leverage.HierarchicalKineticReplicateLayer. rate_term_le_of_rate_constant_bound Leverage/DockingTheoryBridge.lean	L709	Leverage.hierarchical_kinetic_replicate_ inference_bundle_of_rate_constant_margins Leverage/DockingTheoryBridge.lean
L710	Leverage.hierarchical_kinetic_replicate_ inference_bundle_of_rate_constant_margins_ all_datasets Leverage/DockingTheoryBridge.lean	L711	Leverage.microscopic_langevin_explicit_sde_ conditions Leverage/DockingTheoryBridge.lean
L712	Leverage.microscopic_langevin_endpoints_of_ derivation Leverage/DockingTheoryBridge.lean	L713	Leverage.continuous_state_relevance_transport_ iff_of_measurable_encoding_bridge Leverage/DockingTheoryBridge.lean
L714	Leverage.continuous_state_partition_fiber_ image_of_measurable_encoding_bridge Leverage/DockingTheoryBridge.lean	L715	Leverage.continuous_state_srank_transport_of_ measurable_encoding_bridge Leverage/DockingTheoryBridge.lean

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L716	Leverage.continuous_state_energyLowerBound_ transport_of_measurable_encoding_bridge Leverage/DockingTheoryBridge.lean	L717	Leverage.electronicStructureCorrection_ lipschitz Leverage/DockingTheoryBridge.lean
L718	Leverage.electronicStructureCorrection_abs_ error_bound Leverage/DockingTheoryBridge.lean	L719	Leverage.concreteComposedHamiltonian_with_ electronic_realism_lipschitz_bound Leverage/DockingTheoryBridge.lean
L720	Leverage.concreteComposedHamiltonian_with_ electronic_realism_energy_error_bound Leverage/DockingTheoryBridge.lean	L721	Leverage.concreteComposedHamiltonian_with_ electronic_realism_halfGapTransport Leverage/DockingTheoryBridge.lean
L722	Leverage.biomolecularElectronicReference_ shell_error_values Leverage/DockingTheoryBridge.lean	L723	Leverage.biomolecularElectronicReference_ lipschitz_bound Leverage/DockingTheoryBridge.lean
L724	Leverage.biomolecularElectronicReference_ energy_error_bound Leverage/DockingTheoryBridge.lean	L725	Leverage.microscopic_langevin_ergodic_of_ dissipativity Leverage/DockingTheoryBridge.lean
L726	Leverage.continuous_state_transport_bundle_of_ measurable_encoding_bridge Leverage/DockingTheoryBridge.lean	L727	Leverage.biomolecularElectronicReference_ halfGapTransport Leverage/DockingTheoryBridge.lean
L728	Leverage.MolecularHamiltonianThermostatMicroscopicSystem. hamiltonian_energy_matches_realism Leverage/DockingTheoryBridge.lean	L729	Leverage.molecular_hamiltonian_thermostat_to_ first_principles Leverage/DockingTheoryBridge.lean
L730	Leverage.molecular_hamiltonian_thermostat_ explicit_sde_conditions Leverage/DockingTheoryBridge.lean	L731	Leverage.molecular_hamiltonian_thermostat_ endpoints Leverage/DockingTheoryBridge.lean
L732	Leverage.canonical_continuous_path_process_ closure Leverage/DockingTheoryBridge.lean	L733	Leverage.canonical_continuous_path_process_ marginal_recovery Leverage/DockingTheoryBridge.lean
L734	Leverage.QMGroundedElectronicStructureLayer. base_shell_le_tight Leverage/DockingTheoryBridge.lean	L735	Leverage.QMGroundedElectronicStructureLayer. base_error_le_tight Leverage/DockingTheoryBridge.lean
L736	Leverage.qm_grounded_electronic_realism_ lipschitz_bound Leverage/DockingTheoryBridge.lean	L737	Leverage.qm_grounded_electronic_realism_ energy_error_bound Leverage/DockingTheoryBridge.lean
L738	Leverage.unified_chemical_state_dynamics_ observable_transport_bundle Leverage/DockingTheoryBridge.lean	L739	Leverage.UnifiedChemicalStateDynamics. expected_utility_separation_of_margin Leverage/DockingTheoryBridge.lean
L740	Leverage.AbsoluteBindingFreeEnergyCorrections. totalCorrection_abs_le_componentBound Leverage/DockingTheoryBridge.lean	L741	Leverage.AbsoluteBindingFreeEnergyModel. corrected_value Leverage/DockingTheoryBridge.lean
L742	Leverage.AbsoluteBindingFreeEnergyModel.total_ error_bound Leverage/DockingTheoryBridge.lean	L743	Leverage.UnifiedDockingPhysicalModel.thermo_ kinetic_joint_bundle Leverage/DockingTheoryBridge.lean
L744	Leverage.UniversalityOODLayer.ood_error_bound Leverage/DockingTheoryBridge.lean	L745	Leverage.UniversalityOODLayer.uniform_ood_ error_bound Leverage/DockingTheoryBridge.lean
L746	Leverage.hierarchical_chemical_realdatal_ separation_of_required_sample_size Leverage/DockingTheoryBridge.lean	L747	Leverage.hierarchical_kinetic_replicate_ inference_bundle_of_required_replicate_size Leverage/DockingTheoryBridge.lean
L748	Leverage.ProspectiveBenchmarkRecord.empirical_ closure_bundle Leverage/DockingTheoryBridge.lean	L749	Leverage.ProspectiveBenchmarkRecord.blinded_ predictive_reliability Leverage/DockingTheoryBridge.lean
L750	Leverage.molecular_hamiltonian_thermostat_ canonical_path_process_bundle Leverage/DockingTheoryBridge.lean	L751	Leverage.qm_grounded_electronic_realism_ halfGapTransport Leverage/DockingTheoryBridge.lean
L752	Leverage.unified_physical_model_ood_ prospective_bundle Leverage/DockingTheoryBridge.lean	L753	Leverage.hierarchical_required_size_joint_ bundle Leverage/DockingTheoryBridge.lean
L754	Leverage.ConcreteHamiltonianBathLangevinSystem. langevin_endpoints Leverage/DockingTheoryBridge.lean	L755	Leverage.canonical_continuous_path_process_ closure_of_finite_horizon_bridge Leverage/DockingTheoryBridge.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
L756	Leverage.qm_method_specific_realism_transport_bundle Leverage/DockingTheoryBridge.lean	L757	Leverage.ChemicalConditionedTransitionMechanism.unified_dynamics_stationary_bundle Leverage/DockingTheoryBridge.lean
L758	Leverage.AbsoluteFreeEnergyProtocolCalibration.total_error_bound Leverage/DockingTheoryBridge.lean	L759	Leverage.UnifiedPhysicalSimulatorPipeline.thermo_kinetic_bundle Leverage/DockingTheoryBridge.lean
L760	Leverage.OODTransferCalibration.uniform_ood_transfer_bound Leverage/DockingTheoryBridge.lean	L761	Leverage.hierarchical_chemical_reald_data_separation_of_required_sample_size_of_count_order Leverage/DockingTheoryBridge.lean
L762	Leverage.hierarchical_kinetic_replicate_inference_bundle_of_required_replicate_size_of_count_order Leverage/DockingTheoryBridge.lean	L763	Leverage.hierarchical_required_size_joint_bundle_of_count_order Leverage/DockingTheoryBridge.lean
L764	Leverage.GibbsConditionedChemicalMechanismData.unified_dynamics_stationary_bundle Leverage/DockingTheoryBridge.lean	L765	Leverage.qm_protocol_derived_method_specific_realism_transport_bundle Leverage/DockingTheoryBridge.lean
L766	Leverage.rate_div_sqrt_nat_le_of_square_count_bound Leverage/DockingTheoryBridge.lean	L767	Leverage.HierarchicalChemicalRealDataLayer.rate_term_le_of_required_sample_square_bound Leverage/DockingTheoryBridge.lean
L768	Leverage.HierarchicalKineticReplicateLayer.rate_term_le_of_required_replicate_square_bound Leverage/DockingTheoryBridge.lean	L769	Leverage.hierarchical_required_size_joint_bundle_of_square_count_bounds Leverage/DockingTheoryBridge.lean
L770	Leverage.explicit_hamiltonian_bath_elimination_langevin_endpoints Leverage/DockingTheoryBridge.lean	L771	Leverage.explicit_molecular_constant_drift_sde_endpoints Leverage/DockingTheoryBridge.lean
L772	Leverage.concrete_generator_canonical_path_process_closure Leverage/DockingTheoryBridge.lean	L773	Leverage.qm_workflow_specific_realism_transport_bundle Leverage/DockingTheoryBridge.lean
L774	Leverage.ReversibleChemicalTransitionMechanism.unified_dynamics_detailed_balance_bundle Leverage/DockingTheoryBridge.lean	L775	Leverage.TrajectoryAbsoluteFreeEnergyCalibration.total_error_bound Leverage/DockingTheoryBridge.lean
L776	Leverage.UnifiedPhysicalQuantifiedSimulatorPipeline.thermo_kinetic_bundle Leverage/DockingTheoryBridge.lean	L777	Leverage.MechanisticOODTransferModel.uniform_ood_transfer_bound Leverage/DockingTheoryBridge.lean
L778	Leverage.ModelDependentFiniteSampleLaw.rate_term_le_targetMargin Leverage/DockingTheoryBridge.lean	L779	Leverage.hierarchical_required_size_joint_bundle_of_model_dependent_constants Leverage/DockingTheoryBridge.lean
L780	Leverage.hamiltonian_finite_difference_drift_derivation_endpoints Leverage/DockingTheoryBridge.lean	L781	Leverage.realistic_molecular_finite_difference_sde_endpoints Leverage/DockingTheoryBridge.lean
L782	Leverage.generator_coefficients_to_canonical_regularity_closure Leverage/DockingTheoryBridge.lean	L783	Leverage.concrete_qm_workflow_error_analysis_transport_bundle Leverage/DockingTheoryBridge.lean
L784	Leverage.barrier_crossing_reversible_chemical_dynamics_bundle Leverage/DockingTheoryBridge.lean	L785	Leverage.mixing_autocorrelation_trajectory_correction_total_error_bound Leverage/DockingTheoryBridge.lean
L786	Leverage.UnifiedSimulatorErrorAnalysis.unified_controlled_thermo_kinetic_bundle Leverage/DockingTheoryBridge.lean	L787	Leverage.descriptor_calibrated_mechanistic_ood_transfer_bundle Leverage/DockingTheoryBridge.lean
L788	Leverage.model_dependent_minimax_optimality_bundle Leverage/DockingTheoryBridge.lean	L789	Leverage.ConstructiveStochasticAnalysisMultipoleClosure.scope_gap_discharge_bundle Leverage/DockingTheoryBridge.lean
L790	Leverage.ito_wiener_filtration_langevin_endpoints Leverage/DockingTheoryBridge.lean	L791	Leverage.hamiltonian_mori_zwanzig_h_zero_limit_endpoints Leverage/DockingTheoryBridge.lean
L792	Leverage.forcefield_derived_realistic_sde_endpoints Leverage/DockingTheoryBridge.lean	L793	Leverage.generator_pde_estimates_to_canonical_regularity_closure Leverage/DockingTheoryBridge.lean
L794	Leverage.qm_workflow_transport_of_benchmark_summary Leverage/DockingTheoryBridge.lean	L795	Leverage.potential_landscape_barrier_kinetics_bundle Leverage/DockingTheoryBridge.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
L796	Leverage.spectral_gap_concentration_trajectory_bundle Leverage/DockingTheoryBridge.lean	L797	Leverage.spectral_gap_absolute_free_energy_total_error_bound Leverage/DockingTheoryBridge.lean
L798	Leverage.UnifiedSimulatorIntegratorErrorStack.unified_controlled_thermo_kinetic_bundle Leverage/DockingTheoryBridge.lean	L799	Leverage.learned_descriptor_ood_generalization_transfer_bundle Leverage/DockingTheoryBridge.lean
L800	Leverage.estimator_minimax_derivation_bundle Leverage/DockingTheoryBridge.lean	L801	Leverage.extended_physical_model_interface_scope_bundle Leverage/DockingTheoryBridge.lean
L802	Leverage.pre_registered_prospective_benchmark_beats_strong_baselines_bundle Leverage/DockingTheoryBridge.lean	L803	Leverage.independent_replication_outside_team_bundle Leverage/DockingTheoryBridge.lean
L804	Leverage.downstream_campaign_win_bundle Leverage/DockingTheoryBridge.lean	L805	Leverage.external_validation_threeway_integration_bundle Leverage/DockingTheoryBridge.lean
L806	Leverage.extended_physical_model_interface_scope_bundle_of_constructive_closure Leverage/DockingTheoryBridge.lean	L807	Leverage.ContractBoundProspectiveBenchmarkResults.fixed_contract_pre_registered_bundle Leverage/DockingTheoryBridge.lean
L808	Leverage.IndependentReplicationProvenance.outside_team_compute_signed_bundle Leverage/DockingTheoryBridge.lean	L809	Leverage.DownstreamCausalCampaignEvidence.causal_quality_downstream_win_bundle Leverage/DockingTheoryBridge.lean
L810	Leverage.concrete_external_validation_threeway_bundle Leverage/DockingTheoryBridge.lean	L811	Leverage.concrete_external_validation_not_credibly_dismissible Leverage/DockingTheoryBridge.lean
L812	Leverage.langevin_measure_theoretic_endpoint_bundle Leverage/DockingTheoryBridge.lean	L813	Leverage.constructive_ito_wiener_filtration_langevin_endpoints Leverage/DockingTheoryBridge.lean
L814	Leverage.derived_generator_pde_operator_to_canonical_regularity_closure Leverage/DockingTheoryBridge.lean	L815	Leverage.extended_physical_model_interface_scope_bundle_of_microscopic_derivation Leverage/DockingTheoryBridge.lean
L816	Leverage.UnifiedSimulatorIntegratorErrorStack.control_flags_of_numerical_stack Leverage/DockingTheoryBridge.lean	L817	Leverage.attested_concrete_external_validation_threeway_bundle Leverage/DockingTheoryBridge.lean
L818	Leverage.attested_concrete_external_validation_not_credibly_dismissible Leverage/DockingTheoryBridge.lean	L819	Leverage.attested_concrete_artifacts_store_backed_bundle Leverage/DockingTheoryBridge.lean
L820	Leverage.store_backed_attested_concrete_external_validation_threeway_bundle Leverage/DockingTheoryBridge.lean	L821	Leverage.store_backed_attested_concrete_external_validation_not_credibly_dismissible Leverage/DockingTheoryBridge.lean
L822	Leverage.molecularDocking_toDecisionProblem_utility_eq_physicalUtility Leverage/DockingTheoryBridge.lean	L823	Leverage.molecularDocking_sranks_eq_relevant_coordinate_card Leverage/DockingTheoryBridge.lean
L824	Leverage.molecularDocking_sranks_bound_of_excludedProteinAtoms Leverage/DockingTheoryBridge.lean	L825	Leverage.molecularDocking_binarySummary_sranks_1e Leverage/DockingTheoryBridge.lean
L826	Leverage.equilibriumKdOfDrivingEnergy_1e_of_drive_floor Leverage/DockingTheoryBridge.lean	L827	Leverage.molecularDocking_equilibrium_pathRatio_eq_one_of_detailedBalance Leverage/DockingTheoryBridge.lean
L828	Leverage.molecularDocking_equilibrium_freeEnergy_pathRatio_bundle Leverage/DockingTheoryBridge.lean	L829	Leverage.molecularDocking_equilibriumKd_upper_bound_of_rank_lower_bound Leverage/DockingTheoryBridge.lean
L830	Leverage.molecularDocking_contactShell_budget_necessary_of_rank_lower_bound Leverage/DockingTheoryBridge.lean	L831	Leverage.molecularDocking_independent_rank_empirical_prediction_bundle Leverage/DockingTheoryBridge.lean
L832	Leverage.molecularDocking_witness_bundle_of_exactLJ_physics Leverage/DockingTheoryBridge.lean	L833	Leverage.ConcreteDockingPhysicsWitness.derived_witness_bundle Leverage/DockingTheoryBridge.lean
L834	Leverage.driving_free_energy_floor_of_partition_ratio_margin Leverage/DockingTheoryBridge.lean	L835	Leverage.equilibriumKd_upper_bound_of_partition_ratio_drive_floor Leverage/DockingTheoryBridge.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
L836	Leverage.molecularDocking_equilibriumKd_upper_bound_of_rank_lower_bound_from_partition_chain Leverage/DockingTheoryBridge.lean	L837	Leverage.molecularDocking_sranks_interval_of_independent_certificates Leverage/DockingTheoryBridge.lean
L838	Leverage.molecularDocking_independent_certificate_empirical_prediction_bundle Leverage/DockingTheoryBridge.lean	L839	Leverage.DockingRankFalsificationProtocol.not_falsified_iff Leverage/DockingTheoryBridge.lean
L840	Leverage.molecularDockingFalsificationProtocolFromIndependentRankTrajectoryCorrectionEstimator.upper_bound_violation_with_margin_implies_true_violation Leverage/DockingTheoryBridge.lean	L841	Leverage.molecularDockingFalsificationProtocolFromIndependentRankTrajectoryCorrectionEstimator.upper_bound_violation_with_margin_implies_true_violation Leverage/DockingTheoryBridge.lean
L842	Leverage.DockingRankFalsificationProtocol.high_confidence_fail_condition_bundle Leverage/DockingTheoryBridge.lean	L843	Leverage.equilibriumKd_interval_of_drivingEnergy_abs_error Leverage/DockingTheoryBridge.lean
L844	Leverage.equilibriumKd_interval_of_absoluteFreeEnergyModel Leverage/DockingTheoryBridge.lean	L845	Leverage.chemistry_condition_data_backed_kd_interval_bundle Leverage/DockingTheoryBridge.lean
L846	Leverage.PerSystemWitnessDischargeProgram.discharge_all_cases Leverage/DockingTheoryBridge.lean	L847	Leverage.PartitionChainNumericalClosure.kd_upper_bound_bundle Leverage/DockingTheoryBridge.lean
L848	Leverage.ProductionIndependentSranksExtractor.certified_interval_bundle Leverage/DockingTheoryBridge.lean	L849	Leverage.LockedProspectiveFalsificationRun.falsification_bundle Leverage/DockingTheoryBridge.lean
L850	Leverage.AssayNoiseCalibrationModel.high_confidence_call_validity_bundle Leverage/DockingTheoryBridge.lean	L851	Leverage.TargetClassChemistryCalibration.kd_interval_bundle Leverage/DockingTheoryBridge.lean
L852	Leverage.FullPhysicalClosureTargetInstance.bundle Leverage/DockingTheoryBridge.lean	L853	Leverage.ExternalReplicationAtScale.full_pipeline_bundle Leverage/DockingTheoryBridge.lean
L854	Leverage.CertifiedPartitionFunctionComputation.singleStateExact_positive_bundle Leverage/DockingTheoryBridge.lean	L855	Leverage.AbsoluteFreeEnergyProtocolCalibration.zeroOneSample_total_error_bundle Leverage/DockingTheoryBridge.lean
L856	Leverage.ProductionIndependentSranksExtractor.ofUpperSufficiency_interval_bundle Leverage/DockingTheoryBridge.lean	L857	Leverage.FullPhysicalClosureTargetInstance.zeroRankConcreteOfUpperSufficiency_bundle Leverage/DockingTheoryBridge.lean
L858	Leverage.ExternalReplicationAtScale.ofAttestedIndependentReplicationProvenance_bundle Leverage/DockingTheoryBridge.lean	L859	Leverage.concrete_attested_single_target_external_replication_bundle Leverage/DockingTheoryBridge.lean
L860	Leverage.computableBestPoseByUtility_mem_opt Leverage/DockingTheoryBridge.lean	L861	Leverage.RMSDPoeriorModel.successProbability_unit_interval Leverage/DockingTheoryBridge.lean
L862	Leverage.RMSDPoeriorModel.dockingTopK_mass_le_successProbability_of_covered Leverage/DockingTheoryBridge.lean	L863	Leverage.RMSDProbabilityDerivedPoseSolver.bundle Leverage/DockingTheoryBridge.lean
L864	Leverage.computable_pose_solver_and_rmsd_probability_bundle Leverage/DockingTheoryBridge.lean	L865	Leverage.sampledDockingSolverInputFromRawPocketLigand_bundle Leverage/DockingSolverEndpoint.lean
L866	Leverage.DeploymentRMSDCalibration.deployment_contract_implies_benchmark_contract Leverage/DockingSolverEndpoint.lean	L867	Leverage.CanonicalProgramExecutionWitness.refines_solver_result Leverage/DockingSolverEndpoint.lean
L868	Leverage.solveRawPocketLigandBenchmark_bundle Leverage/DockingSolverEndpoint.lean	L869	Leverage.solveRawPocketLigandCanonicalBenchmark_accept_iff Leverage/DockingSolverEndpoint.lean
L870	Leverage.solveRawPocketLigandCanonicalDeployment_accepted_iff Leverage/DockingSolverEndpoint.lean	L871	Leverage.rawCanonicalProgramExecutionWitnessOfSpec_refines_solver_result_canonical Leverage/DockingSolverEndpoint.lean
L872	Leverage.rawPocketLigandCanonicalDefinitiveEndpoint_bundle Leverage/DockingSolverEndpoint.lean	L873	Leverage.runCanonicalSolverProgram_refines_solver_result Leverage/DockingSolverEndpoint.lean
L874	Leverage.solveDefinitiveRawCrossDock_accepted_iff_benchmark_contract Leverage/DockingSolverEndpoint.lean	L875	Leverage.solveDefinitiveRawCrossDock_accepted_iff_deployment_contract Leverage/DockingSolverEndpoint.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
L876	Leverage.solveDefinitiveRawCrossDock_total Leverage/DockingSolverEndpoint.lean	L877	Leverage.runDefinitiveRawCrossDockProgram_ refines_benchmark_accept Leverage/DockingSolverEndpoint.lean
L878	Leverage.solveDefinitiveRawCrossDock_full_ closure_bundle Leverage/DockingSolverEndpoint.lean	L879	Leverage.solveDefinitiveRawCrossDockBenchmark_ accepted_iff_benchmark_contract Leverage/DockingSolverEndpoint.lean
L880	Leverage.solveDefinitiveRawCrossDock_rejected_ iff_not_deployment_contract Leverage/DockingSolverEndpoint.lean	L881	Leverage.definitiveRawCrossDockAcceptanceFlag_ true_iff_deployment_accepted Leverage/DockingSolverEndpoint.lean
L882	Leverage.definitiveRawCrossDockAcceptanceFlag_ false_iff_deployment_rejected Leverage/DockingSolverEndpoint.lean	L883	Leverage.RationalizedAcceptanceKernel. computableAcceptFlag_true_iff Leverage/DockingSolverEndpoint.lean
L884	Leverage.RationalizedAcceptanceKernel. computableAcceptFlag_sound Leverage/DockingSolverEndpoint.lean	L885	Leverage.RationalizedAcceptanceKernel. computableAcceptFlag_refines_benchmark_accept Leverage/DockingSolverEndpoint.lean
L886	Leverage.interpretCanonicalProgramState_ refines_solver_result Leverage/DockingSolverEndpoint.lean	L887	Leverage.interpretDefinitiveRawCrossDockProgram_ eq_run Leverage/DockingSolverEndpoint.lean
L888	Leverage.buildDefinitiveRawCrossDockReport_ runtime_accept_iff_deployment_accepted Leverage/DockingSolverEndpoint.lean	L889	Leverage.definitiveRawCrossDockCompleteLeanBundle Leverage/DockingSolverEndpoint.lean
L890	Leverage.buildDefinitiveRawCrossDockReport_ runtime_reject_iff_deployment_rejected Leverage/DockingSolverEndpoint.lean	L891	Leverage.solveDefinitiveRawCrossDockBenchmarkConstructive_ accepted_iff_kernel_flag Leverage/DockingSolverEndpoint.lean
L892	Leverage.solveDefinitiveRawCrossDockBenchmarkConstructive_ refines_benchmark_accept Leverage/DockingSolverEndpoint.lean	L893	Leverage.solveDefinitiveRawCrossDockConstructive_ refines_deployment_accept Leverage/DockingSolverEndpoint.lean
L894	Leverage.ExactRatArtifactInstantiation. constructive_accepted_iff_benchmark_contract Leverage/DockingSolverEndpoint.lean	L895	Leverage.ExactRatArtifactInstantiation. constructive_accept_refines_certificate_ backend_accepts Leverage/DockingSolverEndpoint.lean
L896	Leverage.solveDefinitiveRawCrossDockBenchmarkDecision_ eq_constructive Leverage/DockingSolverEndpoint.lean	L897	Leverage.solveDefinitiveRawCrossDockDecision_ eq_constructive Leverage/DockingSolverEndpoint.lean
L898	Leverage.solveDefinitiveRawCrossDockBenchmarkDecision_ refines_certificate_backend_accept Leverage/DockingSolverEndpoint.lean	L899	Leverage.solveDefinitiveRawCrossDockDecision_ refines_certificate_backend_accept Leverage/DockingSolverEndpoint.lean
L900	Leverage.solveDefinitiveRawCrossDockBenchmarkDecision_ rejected_iff_kernel_flag_false Leverage/DockingSolverEndpoint.lean	L901	Leverage.solveDefinitiveRawCrossDockBenchmarkCertified_ accepted_iff Leverage/DockingSolverEndpoint.lean
L902	Leverage.solveDefinitiveRawCrossDockBenchmarkCertified_ rejected_iff Leverage/DockingSolverEndpoint.lean	L903	Leverage.solveDefinitiveRawCrossDockCertified_ accepted_iff Leverage/DockingSolverEndpoint.lean
L904	Leverage.solveDefinitiveRawCrossDockCertified_ rejected_iff Leverage/DockingSolverEndpoint.lean	L905	Leverage.SignedRationalizedKernelArtifact. manifest_consistency_bundle Leverage/DockingSolverEndpoint.lean
L906	Leverage.SignedRationalizedKernelArtifact. decision_accept_refines_certificate_backend_ accept Leverage/DockingSolverEndpoint.lean	L907	Leverage.ExactRatArtifactInstantiation. constructive_rejected_refines_certificate_ backend_rejections Leverage/DockingSolverEndpoint.lean
L908	Leverage.SignedExactRatKernelArtifact. benchmark_decision_accepted_iff_benchmark_ contract Leverage/DockingSolverEndpoint.lean	L909	Leverage.SignedExactRatKernelArtifact. decision_rejected_refines_certificate_backend_ rejections Leverage/DockingSolverEndpoint.lean
L910	Leverage.parseSignedArtifactByteEnvelope_ encode Leverage/DockingSolverEndpoint.lean	L911	Leverage.concreteChecksum_parse_verify_end_to_ end Leverage/DockingSolverEndpoint.lean
L912	Leverage.SignedRationalizedKernelArtifact. concreteChecksum_byte_parse_and_verify Leverage/DockingSolverEndpoint.lean	L913	Leverage.RationalizedAcceptanceKernel. rejectionSeparation_not_benchmarkContract Leverage/DockingSolverEndpoint.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
L914	Leverage.RationalizedAcceptanceKernel. rejectionSeparation_flag_false Leverage/DockingSolverEndpoint.lean	L915	Leverage.SignedRationalizedKernelArtifact. strict_rejection_refines_certificate_backend_ rejections Leverage/DockingSolverEndpoint.lean
L916	Leverage.concreteChecksumArtifactSignatureVerifier. verify_iff Leverage/DockingSolverEndpoint.lean	L917	Leverage.definitiveComputableTotalOps_closed_ form Leverage/DockingSolverEndpoint.lean
L918	Leverage.definitiveComputableTotalOps_succFuel Leverage/DockingSolverEndpoint.lean	L919	Leverage.runDefinitiveComputablePipeline_ totalOps_closed_form Leverage/DockingSolverEndpoint.lean
L920	Leverage.branchAndBoundPrune_sound Leverage/DockingSolverEndpoint.lean	L921	Leverage.adaptiveCampaignStopRule_sound Leverage/DockingSolverEndpoint.lean
L922	Leverage.runDefinitiveComputablePipelineBranchAndBound_ prune_sound Leverage/DockingSolverEndpoint.lean	L923	Leverage.definitiveComputablePipeline_ batchFusionJustified Leverage/DockingSolverEndpoint.lean
L924	Leverage.parseSignedArtifactByteEnvelope_cost_ linear_time Leverage/DockingSolverEndpoint.lean	L925	Leverage.parseSignedArtifactByteEnvelope_ encode_cost_exact Leverage/DockingSolverEndpoint.lean
L926	Leverage.cryptographicArtifactSignatureVerifier_ sound Leverage/DockingSolverEndpoint.lean	L927	Leverage.SignedRationalizedKernelArtifact. crypto_byte_parse_and_verify Leverage/DockingSolverEndpoint.lean
L928	Leverage.runDefinitiveComputablePipelineOfSignedArtifact_ parserBytes_exact Leverage/DockingSolverEndpoint.lean	L929	Leverage.definitiveComputableCampaignPairEvaluations_ closed_form Leverage/DockingSolverEndpoint.lean
L930	Leverage.definitiveComputableCampaignPairEvaluations_ succFuel Leverage/DockingSolverEndpoint.lean	L931	Leverage.runDefinitiveComputablePipeline_ campaignPairEvaluations_closed_form Leverage/DockingSolverEndpoint.lean
L932	Leverage.definitiveComputablePipeline_ pairPotentialFusionJustified Leverage/DockingSolverEndpoint.lean	L933	Leverage.SampledDockingSolverInput. canonicalSolverProgramIR_scorerFusionSound Leverage/DockingSolverEndpoint.lean
ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean	PH26	PhysicalComplexity.no_collapse_of_bounded_ budget_pos_cost_exp_lb paper4/DecisionQuotient/Physics/ PhysicalHardness.lean
QT1	DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean	QT2	DecisionProblem.quotientMap_preservesOpt paper4/DecisionQuotient/Quotient.lean
QT3	DecisionProblem.quotient_represents_opt_equiv paper4/DecisionQuotient/Quotient.lean	QT7	DecisionProblem.quotient_has_unique_ factorization paper4/DecisionQuotient/Quotient.lean
SE1	ClaimClosure.SE1 paper4/DecisionQuotient/ClaimClosure.lean	SE2	ClaimClosure.SE2 paper4/DecisionQuotient/ClaimClosure.lean
SE3	ClaimClosure.SE3 paper4/DecisionQuotient/ClaimClosure.lean	SE4	ClaimClosure.SE4 paper4/DecisionQuotient/ClaimClosure.lean
SE5	ClaimClosure.SE5 paper4/DecisionQuotient/ClaimClosure.lean	SE6	ClaimClosure.SE6 paper4/DecisionQuotient/ClaimClosure.lean
W1	Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W2	Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
W3	Physics.integrity_is_centroid paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W4	Physics.wasserstein_bridge paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
WM4	Physics.WolpertMismatch.mismatchNatLowerBound_ pos_of_exists_ne paper4/DecisionQuotient/Physics/ WolpertMismatch.lean	WM6	Physics.WolpertDecomposition.effective_model_ strictly_exceeds_landauer_of_distribution_ mismatch paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean
WP2	Physics.WolpertDecomposition.landauer_floor_ plus_decomposition_lower_bound paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean	WP6	Physics.WolpertDecomposition.effective_model_ strictly_exceeds_landauer_of_either_cited_ component paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
WP7	Physics.WolpertDecomposition.landauer_floor_plus_structural_resource_lower_bound paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean	WP8	Physics.WolpertDecomposition.energy_lower_bound_increases_by_structural_resource paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean
WP9	Physics.WolpertDecomposition.physical_grounding_bundle_with_wolpert_decomposition paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean	WR6	Physics.WolpertResidual.discreteResidualNatLowerBound_pos_of_asymmetry_or_oneway paper4/DecisionQuotient/Physics/ WolpertResidual.lean
WR7	Physics.WolpertDecomposition.stopping_time_residual_of_discrete_edge_split paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean	WR10	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_finite_discrete_witness paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean
WR11	Physics.WolpertResidual.binaryEncodedResidualNatLowerBound_eq_one paper4/DecisionQuotient/Physics/ WolpertResidual.lean	WR12	Physics.WolpertDecomposition.effective_model_ge_landauer_plus_one_of_binary_encoded_residual_example paper4/DecisionQuotient/Physics/ WolpertDecomposition.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 5.323: Concrete Bond-Family Holonomic Floor	L396, L398, L397
Corollary 4.11: Bounded Contact-Shell Low-Rank Regime	L142
Corollary 4.7: Bounded-Pocket Low-Rank Regime	L65
Corollary 5.18: Decision-Entropy Bound from Spacetime and Energy Budget	IT3, BA1, BA2, BA5, BA6, EI1, L43
Corollary 4.5: Checking Time Lower Bound	L70
Corollary 5.19: Independent Composition Budget Law	L17, L19, BA1, BA2, BA7, BA5, BA6, L43
Corollary 5.322: RATTLE Holonomic-Constraint Landauer Floor	L60, L61, L62
Corollary 5.325: Jacobian-Rank Holonomic Floor for Nonlinear Constraints	L405, L407, L403, L404, L406
Corollary 5.307: Logical Basement and Realized Dissipation	L140, BA5, BA6, WM6, WP6, WR10, L43
Corollary 5.6: Unique Minimum-Cost Regime	BA8, L54, L55
Corollary 4.4: No Sound Checker Below Witness Budget	L71
Corollary 3.38: Higher-Rank Regime	L46
Corollary 3.37: Rank-One Regime	L51
Proposition 5.21: Two-Level Atomic Realization	AC1, AC3, AC4
Proposition 5.318: Repeated Binary Mismatch Work Law	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.317: Binary Mismatch Strengthens the Energy-Information Coefficient	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.316: Explicit Binary Mismatch Example	BA5, BA6, WP2, WM4, L43
Proposition 5.311: Repeated Residual-Example Work Law	IT3, BA5, BA6, WR12, WR11, L43
Proposition 5.310: Explicit Two-State Residual Example	IT3, BA5, BA6, WR12, WR11, L43
Proposition 2.10: Binding as Exact Resolution	BA3, BA4, BA5

Paper claim	Lean handle
Proposition 2.5: Bounded Region	BA1
Proposition 5.314: Canonical Wolpert Grounding Bundle	BA5, BA6, WP9, L43
Definition 2.3: Bounded Decision System	L17, L19
Proposition 5.312: Unified Energy–Information Hierarchy	IT3, BA5, BA6, WR12, WP2, WM4, WR11, L43
Proposition 3.44: Finite Compression-Relation Bridge	L57, L58
Proposition 5.309: Finite Discrete Residual Witness	WR10, WR7, WR6
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Proposition 5.320: Finite Lifetime Throughput Bound	SE5, IT3, L43
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Proposition 5.315: Strict Canonical Energy Above the Landauer Floor	BA5, BA6, WP6, L43
Proposition 5.308: Theorem-Level Strict Overhead Branches	BA5, BA6, WM6, WP6, WR10, L43
Proposition 5.313: Structural-Resource Overhead	BA5, BA6, WP8, WP7, L43
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Paper claim	Lean handle
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Paper claim	Lean handle
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Auto summary: mapped 453/453 (full=453, derived=0, unmapped=0).

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thm:	unspecified	-	L752
unified-physical-ood-prospective-bundle			
thm:	unspecified	-	L786
unified-simulator-error-analysis-controlled-thermo-kinetic			
thm:	unspecified	-	L759
unified-simulator-kinetic-derivation			
thm:	unspecified	-	L743
unified-thermo-kinetic-physical-model			
thm:	unspecified	-	L744, L745
universality-ood-calibration-bounds			
thm:	unspecified	-	L856
upper-only-independent-srank-extractor-bundle			
thm:	unspecified	-	L86
velocity-verlet-volume			
thm:	unspecified	-	L855
zero-correction-calibration-bundle			

Auto summary: indexed 453 claims by hardness profile (unspecified=453).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- Leverage/BridgeToDQ.lean
- Leverage/FiveWayEquivalence.lean
- Leverage/Physical.lean

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)